

a

Dropout

	PCA	PcaGrid	PcaHubert	PCP	K's tau	Winsor	Quad	Ball	Shell	LR
Dropout midpoint	5	10	9	1	8	2	4	7	3	6
1	NR	NR	NR	NR	NR	NR	NR	NR	NR	NR
2	4	10	9	2	1	6	5	3	7	8
3	3	10	8	9	1	7	4	6	5	2
4	1	10	8	9	4	2	5	3	6	7

b

Synthetic doublets

	PCA	PcaGrid	PcaHubert	PCP	K's tau	Winsor	Quad	Ball	Shell	LR
Doublet fraction	4	10	9	1	8	2	6	5	3	7
2%	5	10	9	2	8	4	3	7	1	6
5%	3	10	9	1	8	6	7	5	2	4
8%	3	10	9	1	8	2	4	7	5	6
10%	2	10	9	1	8	7	6	5	3	4
15%	2	10	9	1	8	7	6	5	3	4

c

Gene-subset mean shift

	PCA	PcaGrid	PcaHubert	PCP	K's tau	Winsor	Quad	Ball	Shell	LR
Shifted-cell fraction	4	10	9	1	5	8	6	7	3	2
2%	6	10	9	1	5	8	4	7	3	2
4%	4	10	9	1	2	8	6	7	3	5
6%	4	10	9	1	6	8	7	2	5	3
8%	5	10	9	1	2	8	7	6	4	3
10%	5	10	9	1	2	8	7	6	4	3

d

Gene-subset mean shift

	PCA	PcaGrid	PcaHubert	PCP	K's tau	Winsor	Quad	Ball	Shell	LR
Shift magnitude	3	10	9	1	7	8	6	5	2	4
1.5	4	10	9	1	5	8	6	7	3	2
2	6	10	9	1	2	8	7	5	3	4
3	6	10	9	1	8	7	5	4	3	2
4	6	10	9	1	8	7	5	4	3	2
5	6	10	9	1	8	7	5	4	3	2

